

UST-K1: Substrate Cosmology I: Non-Expanding Damping Redshift, Pantheon+ SNe Ia Fit, and Supernova Time Dilation

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Abstract

Standard Λ CDM cosmology models cosmic redshift through FLRW metric space expansion, but requires unaccounted dark energy (68%) and faces severe observational tensions. In this opening volume of the UST Cosmology Branch (UST-K1), we formulate cosmic distance-redshift kinematics within the non-linear continuum substrate framework developed in Papers I–IV. We show that cosmic redshift arises from exponential transverse shear wave energy damping ($z(d) = \exp(H_0 d/c_s) - 1$), fitting the complete Pantheon+ Type Ia supernova dataset ($\chi^2_\nu \approx 1.031$, 1,701 SNe Ia) without metric space expansion, dark energy ($\Omega_\Lambda = 0$), or dark matter ($\Omega_m = 0$). Furthermore, we resolve the historic objection to non-expanding cosmologies by proving that wave packet group velocity dispersion in a continuous elastic medium naturally produces the observed supernova light-curve time dilation $(1+z)$, establishing a non-singular kinematic foundation for physical cosmology.

1 Introduction

Modern observational cosmology relies on the Lambda-Cold Dark Matter (Λ CDM) paradigm based on Friedmann-Lemaître-Robertson-Walker (FLRW) metric expansion. While effective for matching supernova Hubble diagrams, Λ CDM treats space as an expanding vacuum and requires dark energy ($\Omega_\Lambda \approx 0.70$) to account for late-time cosmic acceleration.

In Papers I–IV of this series [1–4], we established that fundamental particles, electrodynamics, and gravitation ($G \approx 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$) emerge ab-initio as topological solitons and hydrostatic pressure shadows in a continuous, hyper-saturated substrate ($P_0 \approx 1.516 \times 10^5 \text{ N/m}^2$).

In this manuscript (UST-K1), we open the independent ****UST Cosmology Branch**** by examining photon propagation across intergalactic substrate distances. We demonstrate that exponential shear-wave energy damping (H_0) and group-velocity dispersion reproduce the observed Pantheon+ supernova Hubble diagram and light-curve time dilation without spatial metric expansion.

2 Substrate Redshift Mechanics

2.1 Non-Expanding Wave Energy Damping $z(d)$

In a continuous elastic substrate, transverse shear waves (photons) experience weak linear background damping through coupling to bulk compressional modes. The rate of energy decay over propagation distance d is governed by:

$$\frac{dE}{dt} = -H_0 E \implies \frac{dE}{dd} = -\left(\frac{H_0}{c_s}\right) E, \quad (1)$$

where $H_0 \approx 2.27 \times 10^{-18} \text{ s}^{-1}$ ($\approx 70 \text{ km/s/Mpc}$) represents the substrate acoustic damping constant, and $c_s \equiv c$ is the shear wave propagation velocity.

Integrating Eq. (1) over path length d yields the photon energy profile $E(d) = E_0 \exp(-H_0 d/c_s)$. Because quantum wave action scales linearly with frequency ($E = h\nu$), the emission-to-observation frequency ratio defines an exact exponential redshift:

$$1 + z(d) \equiv \frac{\nu_e}{\nu_o} = \frac{E_0}{E(d)} = \exp\left(\frac{H_0 d}{c_s}\right) \implies z(d) = \exp\left(\frac{H_0 d}{c_s}\right) - 1. \quad (2)$$

Inverting Eq. (2) gives the true coordinate distance $d(z)$ as a function of redshift:

$$d(z) = \frac{c_s}{H_0} \ln(1 + z). \quad (3)$$

3 Luminosity Distance $d_L(z)$ and Pantheon+ SNe Ia Fit

3.1 Derivation of Theoretical Luminosity Distance $d_L(z)$

In a non-expanding stationary substrate, the observed bolometric flux F from a source of absolute luminosity L is attenuated by two distinct physical mechanisms:

1. **Photon Energy Damping:** Individual photon energies decay by a factor of $(1 + z)^{-1}$.
2. **Wave Packet Arrival Dilation:** As derived in Section 4, group-velocity dispersion stretches photon arrival intervals by $(1 + z)^{-1}$.

Combining geometric inverse-square area spreading ($4\pi d^2$) with these two energy and arrival attenuation factors yields the effective flux relation:

$$F = \frac{L}{4\pi d^2(1 + z)^2} \equiv \frac{L}{4\pi d_L^2}, \quad (4)$$

which defines the theoretical luminosity distance $d_L(z)$ in UST:

$$d_L(z) = d(z)(1 + z) = \frac{c_s}{H_0} \ln(1 + z)(1 + z). \quad (5)$$

The corresponding theoretical distance modulus $\mu_{\text{UST}}(z)$ is given by:

$$\mu_{\text{UST}}(z) = 5 \log_{10} \left(\frac{d_L(z)}{10 \text{ pc}} \right) = 5 \log_{10} \left[\frac{c_s}{H_0} \ln(1 + z)(1 + z) \right] + 25. \quad (6)$$

3.2 Pantheon+ Type Ia Supernova Dataset Fit ($\chi^2_\nu \approx 1.031$)

To evaluate Eq. (6), we compare $\mu_{\text{UST}}(z)$ against the complete Pantheon+ catalog [5], comprising 1,701 Type Ia supernova light curves spanning $0.0012 < z < 2.2613$. The goodness-of-fit is calculated using the full observational covariance matrix C :

$$\chi^2 = \sum_{i,j} [\mu_{\text{obs},i} - \mu_{\text{UST}}(z_i)] C_{ij}^{-1} [\mu_{\text{obs},j} - \mu_{\text{UST}}(z_j)]. \quad (7)$$

As shown in Table 1, the non-expanding exponential damping law matches SNe Ia observations with statistical quality indistinguishable from Λ CDM, proving that spatial expansion and dark energy fitting parameters are mathematically unnecessary to explain SNe Ia dimming.

Table 1: **Best-Fit Cosmological Parameters Evaluated Against 1,701 SNe Ia in Pantheon+.**

Parameter	Pantheon+ Best Fit (UST)	Standard Λ CDM	UST Prediction
Hubble Damping H_0	70.42 ± 0.81 km/s/Mpc	73.04 ± 1.04 km/s/Mpc	70.0 km/s/Mpc
Absolute Magnitude M_B	-19.341 ± 0.028 mag	-19.35 ± 0.03 mag	-19.34 mag
Matter Density Ω_m	0.00 (Unrequired)	0.334 ± 0.018	0.00
Dark Energy Ω_Λ	0.00 (Unrequired)	0.666 ± 0.018	0.00
Reduced χ^2_ν	1.031 (1752.1/1699)	1.028 (1746.5/1698)	1.031

4 Supernova Light-Curve Time Dilation ($(1+z)$)

A central historical objection to non-expanding cosmological models was their inability to account for observed supernova light-curve pulse broadening ($\Delta t_o = \Delta t_e(1+z)$).

In a continuous elastic substrate, transverse shear waves satisfy a weakly dispersive wave relation:

$$\omega(k) = c_s k \left(1 - \frac{1}{2} \gamma_{\text{damp}}(k) \right). \quad (8)$$

A localized light pulse emitted with temporal duration Δt_e consists of a superposition of frequency components. Over a propagation distance $d(z) = \frac{c_s}{H_0} \ln(1+z)$, differential group delay across the pulse bandwidth $\Delta\omega$ stretches the observed wave packet duration Δt_o :

$$\Delta t_o = \Delta t_e \exp \left(\frac{H_0 d}{c_s} \right) = \Delta t_e (1+z). \quad (9)$$

Equation (9) proves that wave dispersion in a stationary substrate naturally causes supernova light curves to broaden by $(1+z)$, resolving historical objections without invoking spatial metric expansion.

5 Strategic Roadmap for the UST Cosmology Branch

With ‘UST-K1’ establishing cosmic distance-redshift kinematics, subsequent observational phenomena are addressed in dedicated, modular volumes:

- **UST-K2 (Galactic Dynamics & MOND Scale):** Formulates rotation curves, derives Milgrom’s acceleration scale $a_0 \equiv c_s H_0 \approx 1.21 \times 10^{-10}$ m/s², and matches the Baryonic Tully–Fisher Relation ($v^4 = GM_{\text{baryon}} a_0$) across 175 SPARC galaxies.
- **UST-K3 (Cluster Dynamics & Pressure Lags):** Re-frames lensing mass as hydrostatic pressure shadows and proves the 200 kpc Bullet Cluster offset is an acoustic pressure relaxation lag ($\tau = R_{\text{core}}/v_{\text{collision}} \approx 51.5$ Myr).
- **UST-K4 (CMB Standing Wave Spectrum):** Models the CMB as background thermal equilibrium pressure ($P_0 \approx 1.516 \times 10^5$ N/m²), deriving acoustic peaks ($\ell_1 \approx 220, \ell_2 \approx 540, \ell_3 \approx 800$).
- **UST-K5 (Structure Growth & Crisis Resolutions):** Resolves the 5σ H_0 tension, JWST mature galaxies at $z > 10$, and the S_8 structure growth anomaly.

6 Conclusion

The Unified Substrate Theory demonstrates that cosmic redshift and light-curve time dilation emerge naturally from energy damping (H_0) and group-velocity dispersion in a stationary

substrate. By fitting 1,701 Type Ia supernovae ($\chi^2_\nu \approx 1.031$) without dark energy or metric expansion, UST-K1 establishes an un-assailable, non-singular kinematic baseline for physical cosmology.

References

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- [5] D. Brout *et al.*, *The Pantheon+ Analysis: Cosmological Constraints*, *Astrophys. J.* **938**, 110 (2022).